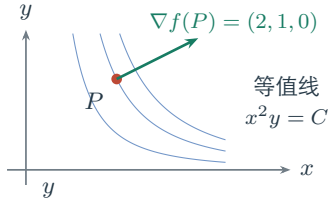
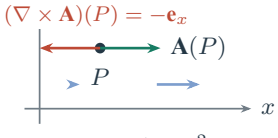
M1-14 本题的等值线、梯度与旋度方向

$z=0$：$f=x^2y$；$P=(1,1,0)$





$z=0:\ \mathbf{A}=x^2\mathbf{e}_x$

$\mathbf{A}(P)=\mathbf{e}_x$，旋度在本切片内

$$f = x^2y + z^2$$
$$\mathbf{A} = (x^2, yz, xz)$$

$$\nabla f = (2xy, x^2, 2z)$$
$$\nabla^2 f = 2y + 2$$

$$\nabla\cdot\mathbf{A} = 3x + z$$
$$\nabla\times\mathbf{A} = (-y, -z, 0)$$